The objective for this project is for you to experiment with building and evaluation a foundation model for representation learning using a self-supervised learning (SSL) approach. To allow for sufficient time for your experiments, we will start with a simple backbone model (resnet-18) and a dataset of small images (CIFAR-10 with 32x32 images). Once you have this baseline model working, you can then try with variations.

Basics of the SimCLR method using contrastive learning:

- As mentioned in the class, the figure shown above illustrates the basic concept of SimCLR. We focus on the first version of SimCLR; the original paper is available at https://arxiv.org/abs/2002.05709.
- The network in SimCLR training include the backbone (in our case, a modified resnet-18, excluding the fully connected layer at the end) and a projector head (a two-layer multi-layer perceptron in the original paper).
- For a batch size of N source images, two independent augmentations are generated, so the effective batch size is 2N. Multiple transforms are used, commonly including random crop, horizontal flip, color jitter, and random grayscale. A normalization step at the end is required.
- The loss function to be used is called **NT-Xent** loss in the original paper. The idea is explained below:
  - Consider an image $\boldsymbol{x}$ in the batch (after augmentation). There are 2N–1 other images in the batch. The task is to identify the mate (the other image in the batch from the same source) of $\boldsymbol{x}$.
  - For all the images in a batch, normalize their output vectors (after the projector head) and then compute their pairwise cosine similarity. For each $\boldsymbol{x}$, you get 2N–1 similarities with the other images.
  - Divide the similarities with a "**temperature**" parameter, then compute their softmax values. The loss for $\boldsymbol{x}$ is the **negative log** of the softmax value of the mate. (This is equivalent to cross-entropy loss if we treat "finding the mate of $\boldsymbol{x}$" as a classification problem.)
  - The loss for a batch is the average loss for all the images in the batch.
- Once the SSL training is finished, we discard the projector head. The "representation" we want is the output of the backbone (for resnet-18, that is after the final global-average-pooling and flatten layers).
- For SimCLR, the batch size affects the training more than in regular supervised learning. A larger batch size provides more variety of negative samples, allowing more effective learning.

More notes about SSL training and evaluation:

- The standard evaluation is to attach a linear (fully connected) layer after the encoder (backbone) output to do a classification task. This method is called **linear probing**: You freeze the weights of the encoder and train only the final linear layer for the classification problem. This provides an indication of the quality of the representation learned.
- The loss curve during training, while decreasing over time, is not a good indicator of SSL learning performance. This is because the loss is not the same as the objective. One common approach to observe the actual quality of the learned representation during training is **kNN monitor**: Do k-nearest-neighbor classification of the test set using current representations of the training set. (You use the encoder outputs, not the projector outputs, for this purpose.) You should see that the kNN accuracy grows during training.

Practical notes about implementation:

- Note: There should be ample open-source samples of SimCLR that you can reference. You can also use AI to guide you through the implementation. Just make sure to mention them in your report.
- All the models are to be trained from scratch (NO pretrained weights).
- To use resnet-18 for CIFAR-10, you need to make change to the first convolution layer to avoid too much downsampling in the beginning. The standard practice:
  - Change its kernel size to 3x3 (7x7 in original resnet) with stride=1 and padding=1.
  - Change the max-pooling layer after the first convolution layer to an identity layer.

- Some settings and hyperparameters to start with (you can modify them in your experiments):
  - Adam optimizer
  - Learning rate: 3e-4
  - Weight decay: 1e-6
  - Temperature: 0.5
  - Projector head: 512 inputs (from backbone) → 512 hidden nodes → 128 outputs
  - Training time: 200 epochs.
  - k=20 (for knn monitor)
  - Do knn monitor for every 5 epochs (this takes some time as well)
  - Linear probing: (fixed) Adam optimizer, 1e-3 learning rate, 1e-6 weight decay, same batch size as SSL training, 100 epochs, full training set used.
- Allowed batch size depends on the VRAM available; you can try 64 for low-end graphics cards, 128 or 256 for mid-range ones, 512 or 1024 for better ones.

Experiments and Analysis:

- Note: The top two points are required. The rest in the list are optional ones that you can explore.
- (Required) Start with the baseline model and report the learning curves (loss and knn-monitor accuracy) during training, as well as the final linear probing results.
- (Required) Train a model with supervised learning (SL) from scratch. You use the same backbone without the projector, and attach a classification head (512 to 10) to the backbone. Compare the accuracy on test set with that of the SSL model (i.e., the linear probing results). How are they different?
- A real lower-bound baseline: Randomly initialize and freeze the weights of the backbone (i.e., the backbone is never trained), and then do linear probing. Compare with the SSL and SL models above.
- For ablation study, the following are more critical for SSL:
  - The types and strengths of augmentation.
  - The choice of temperature. What happens to the learning curves (loss and knn-monitor accuracy) if you use a high temperature (such as 5) or a low temperature (such as 0.1)? Are the changes of the two learning curves consistent?
  - The batch size. From the largest batch size that you can train your model successfully (such as 512), re-do the training with progressively smaller batch sizes (such as 256, 128, 64, 32), and observe the differences in the learning curves.
- You can also try to use no projector head and compute the loss directly using the encoder (backbone) outputs (dim = 512) during training. See what happens with the eventual performance.
- You can also try to use the projector head outputs as the representations (dim = 128) during evaluation. See what happens with the eventual performance.
- A very important property of a foundation model is how well the representation work with new datasets:
  - To do: Freeze the backbone, and do linear probing on a different dataset.
  - Compare the results using the SSL, SL, and baseline models.
  - Options of "other" datasets include CIFAR-100, STL-10, Flowers-102, Food-101, and even the one you created in the first project.
  - Remember to resize (to 32x32) and normalize the input images for this part of experiments.
- For meaningful comparison, the settings for linear probing have to be the same for all experiments, including those involving other datasets.

Discussion:

- Important: Don't just let AI create the discussion for you. Put your thoughts into all the observations and see whether you do or don't understand them.
- Based on your experiments, are the results and observed behaviors what you expect?
- Discuss factors that affect the results, including dataset characteristics.
- Describe experiments that you would do if there were more time available.
- Indicate what you have learned from the experiments as well as your remaining questions.

Submission:

- A report in PDF format (maximum 10 pages single-spaced). It should contain the following sections:
  - A brief statement of you research question and motivation in plain text.
  - Description of the methods used. Provide references for public libraries, open-source codes, and/or AI tools used.
  - Description of your experiments, include evaluation results and examples.
  - A "discussion" section as described above.
  - List of references.
  - Include your program code as an appendix (not counting toward the 10-page limit), starting from a separate page. The code listing should be well-organized and contain comments that help the readers understand your code.
- Additional notes about the report:
  - Submit your report through E3. Late submissions are accepted for up to 5 days, with a deduction of 10% per day.
  - Make sure your student ID is within the filename as well as on the first page.
  - For the description of methods: You should expect the reader to read this part of your report without the code listing. Do NOT write your report like a documentation of your program.
  - Experimental results should come with sufficient description/summarization and discussions. When you list your results, always ask yourself this question: "What message do I want my reader to learn from these numbers?" And then put your answer to text.
  - Presentation of results: Use tables formatted in you report, NOT screenshots. Use charts or plots when appropriate.
  - Legibility is VERY important. Use at least size-12 fonts for your main text and at least size-10 fonts in tables and figures. DO NOT try to cram lots of material into the 10-page limit by using smaller fonts and/or small plots, which will actually hurt your grade.